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IS6052: Predictive Analytics Assignment

Assignment Type: Predictive Analytics (Group Project)

Percentage of Module Mark: 50%

Due Date: on or before Thursday 26th November 2025

Assignment Abstract:

Omni Company is a large subscription-based service provider operating in the telecommunications and digital services industry. It offers internet packages, streaming bundles, premium device options, and customer loyalty programs to millions of subscribers across multiple regions.

In recent years, Omni has faced significant competitive pressure due to:

- The entry of aggressive low-cost competitors.
- Rapid technological changes in the telecom and streaming sectors.
- Shifting consumer habits (e.g., bundling entertainment and internet services).
- Customer dissatisfaction arising from billing disputes, service reliability issues, and competing promotional offers.

In the subscription-based business model, **customer churn**, when a customer stops using a service, directly impacts revenue and profitability. Retaining existing customers is significantly cheaper than acquiring new ones, and high churn rates can damage both financial performance and brand reputation.

Industry reports indicate that telecom and streaming service providers in competitive markets like Omni's face churn rates between 15% and 30% annually. Predicting churn accurately allows companies to:

- Identify high-risk customers before they leave.
- Tailor retention campaigns (e.g., personalized offers).
- Improve service quality in the most churn-prone customer segments.

Omni has decided to invest in predictive analytics to build a Customer Churn Prediction Model. The aim is to use historical customer data to identify the factors most strongly associated with churn, and to design interventions to reduce customer losses. The question Omni Company asks is: Which customers are most likely to churn, and what are the main factors driving churn at Omni Company?

Your model should not only predict churn accurately but also provide insights into the reasons behind customer departures, enabling the business to take proactive retention actions.



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Omni has provided two datasets for this project, both in Excel format:

1. **Main Churn Dataset** (omni_churn_data.csv)
 - Contains pre-churn customer information: demographics, service details, contract terms, usage metrics, billing information, and churn status.
 - This dataset must be used for churn prediction modelling.

The excel file include 35 columns such as:

- **Customer Profile & Demographics**

Attribute	Meaning
CustomerID	Unique number assigned to each customer.
Age	Customer's age in years.
Gender	Customer's gender.
IncomeTier	Customer's approximate income level
Region	Geographic region where the customer lives.
Education	Highest level of education completed.
CityTier	Classification of the city/area into market tiers based on size, economic activity, and competition.
CustomerSegment	This is an internal classification of the customer based on value, usage, or loyalty.

- **Account & Contract Information**

Attribute	Meaning
ContractLength	Length of the current subscription agreement.
PlanType	Type of service plan: Basic, Standard, Premium, or Ultra.
PaymentMethod	How the customer pays (Credit Card, Bank Transfer, Electronic Check, Mailed Check, Crypto).



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TenureMonths	Total number of months the customer has been with Omni Company.
ContractAutoRenew	Whether the contract renews automatically.
AutoPay	Whether payments are automatically charged.
PaymentDelinquencyStatus	This shows the customer's current billing status in relation to overdue payments. For example: possible values in your dataset: "Current" → No overdue payments. "30+" → Payment overdue by 30–59 days.
Paperless	Whether the customer receives bills online instead of by mail.

- Usage & Spend

Attribute	Meaning
MonthlyCharges	Average monthly bill amount.
TotalCharges	Total amount billed to the customer over their entire relationship.
LoginsLastMonth	Number of times the customer logged into their account in the past month.
RFMScore	Composite engagement score based on Recency, Frequency, and Monetary value.
UsageChangePct	Percentage change in service usage compared to a previous period.
CompetitorIndex	Measure of competition intensity in the customer's area (higher = more competitors).

- Engagement & Support

Attribute	Clear Meaning
TicketsOpened	Number of customer support tickets opened.
TicketsResolutionTime	Average hours taken to resolve support requests.
SupportChannelPreferred	Customer's preferred way to contact support (Phone, Email, Chat, etc.).
ComplaintCategory	Main category of the customer's complaint (Billing, Technical, Service, etc.).



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- **Products & Bundles**

Attribute	Meaning
FamilyPlan	Whether the customer has a multi-user (family) plan.
AddOnBundle	Extra services subscribed (Sports, Movies, Security, etc.).
DiscountType	Type of discount applied (Loyalty, Bundle, Promo, etc.).
PromoCodeUsed	Marketing promo code applied at signup or renewal. It's the marketing promotion code the customer applied when they signed up, renewed, or upgraded their plan.

- **Acquisition & Device Preferences**

Attribute	Meaning
ReferralSource	How the customer first learned about OmniConnect (Friend, Search, Affiliate, etc.).
ChannelPreferred	Preferred signup channel (App, Web, In-store, etc.).
DeviceType	Main device used to access OmniConnect services (Mobile, Desktop, Tablet, etc.; includes typos).
DeviceOS	Operating system of the device used (Android, iOS, Windows, etc.).

2. Return Dataset (omni_return_data.csv)

- Contains **post-churn** customer information for customers who returned after leaving:
 - ReturnedCustomer, ReturnAfterMonths, ReturnReason, WasContactedForWinback, WinbackOfferType, CompetitorUsedDuringChurn, ReturnedContractType, ReturnedAddOnBundle, ReturnedPaymentMethod, CustomerSatisfactionPostReturn, ReturnedLoyaltyPointsBalance.
- **These columns (Return Dataset) are not to be used for churn prediction, as they represent information only available after the churn event.**
- **They can be used separately for a Return Analysis project, but not for the churn prediction modelling in this assignment.**



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Important Instruction

Before modelling churn:

- If you choose to merge the two datasets, you **must** drop all post-churn variables from the final modelling dataset to avoid data leakage.
- You will be **penalized** if post-churn features are included in churn prediction models. You may, however, use post-churn attributes for other purposes, such as finding outliers or extracting important information that can help with churn analysis.

Dataset Size

- Number of customers: 50,000 records.
- Target variable: Churn (0 = No churn, 1 = Churn).
- Data type mix: categorical and numeric attributes.
- Files provided:
 - omni_churn_data.xlsx (pre-churn data, for prediction modelling)
 - omni_return_data.xlsx (post-churn data, remove for Churn Modelling)

Assignment Tasks

- **Dataset Description and Initial Exploration (15%):**
 - Provide a detailed description of the dataset, including its nature (e.g., demographic, service usage, account history and so on) and the type of information it contains.
 - Specify the dataset size (number of records and variables), the target variable (Churn), and the data types present (e.g., numerical, categorical, nominal/multinomial).
 - Conduct an initial exploratory data analysis (EDA) to examine the distribution of the target variable, summarise descriptive statistics for numerical features, and review the frequency counts for categorical variables.
 - Identify class imbalance in the target variable, if any, and explore initial patterns or relationships between features and churn.
 - Present EDA findings using appropriate visualisations such as histograms, bar



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charts, boxplots, and heatmaps. Summarise key insights and potential areas of interest for further analysis.

- **Data Preparation and Feature Engineering (10%):**
 - Cleanse and prepare the dataset for analysis by handling missing and incomplete values, resolving inconsistent categorical entries, and standardising formats where necessary.
 - Encode categorical and binary variables appropriately to make them suitable for modelling.
 - Perform feature engineering to create new meaningful variables that may improve model performance. Examples include transformations, derived metrics (e.g., total charges per month), feature interactions, or aggregations.
 - Document all preprocessing and feature engineering steps, explaining the rationale behind each and how they are expected to improve the models.
- **Outlier Detection and Handling (15%):**
 - Identify potential outliers in numerical features and in relevant categorical–numerical combinations using appropriate methods (e.g., interquartile range (IQR), z-score thresholds, domain-specific business rules).
 - Discuss the chosen detection method and why it is suitable for this dataset.
 - Handle outliers appropriately (e.g., removal, capping, transformation, and so, if possible) and justify your approach in the context of churn prediction.
 - Evaluate the impact of outlier handling on dataset quality and model performance, and report any changes made to the dataset.
- **Predictive Analysis (25%):**
 - Apply a range of predictive modelling techniques, including at least: Logistic Regression, SVM, ANN, Decision Tree, Random Forest, and Gradient Boosting/XGBoost and so on. Justify your choice of each technique and explain its relevance to the churn prediction problem.
 - Implement feature selection and apply multiple interpretability approaches, including:
 - Random Forest feature importance
 - SHAP (Shapley Additive Explanations) for global and local interpretation



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- LIME (Local Interpretable Model-agnostic Explanations) for local predictions
 - LOO (Leave-One-Out) feature importance for model-agnostic impact estimation.
- Perform hyperparameter tuning for each model using appropriate methods and document the best parameters found.
- Use cross-validation to ensure model robustness.
- Apply additional techniques, if possible, to improve accuracy or interpretability, such as feature selection, Principal Component Analysis (PCA), Lasso regularisation, or ensemble methods.
- Document and compare how each interpretability method ranks the importance of features and explain any differences in results.
- Compare the performance of models trained on all available features with those trained on a selected subset of important features.
- **Results, Evaluation, and Discussion (25%):**
 - Present detailed model performance results for each method, including metrics such as Accuracy, Precision, Recall, F1-score, and ROC-AUC.
 - Include and discuss confusion matrices for all key models to show true positives, false positives, true negatives, and false negatives, and interpret what these mean for churn prediction in practical terms.
 - Provide additional visualisations to support your analysis, which may include:
 - ROC curves for each model
 - Precision-Recall curves
 - Bar charts of feature importance
 - Correlation heatmaps to show feature interrelationships.
- Critically discuss the effectiveness of each method in predicting churn, considering both predictive power and interpretability.
 - Evaluate the application areas, strengths, and limitations of each modelling technique in the context of your dataset, addressing the following:
 - a) Is the method capable of capturing complex, non-linear relationships?
 - b) Is it easy to interpret and explain to a business audience?
 - c) Does it scale efficiently for large datasets?



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- d) How easy is it to communicate results to non-technical stakeholders?
- e) Does it perform consistently across different scenarios, or is it limited to certain types of data relationships?
- f) Does it require extensive preprocessing, or can it handle raw features effectively?
- Discuss how your preprocessing steps, feature engineering, and outlier handling influenced model performance, and compare results for models with and without these steps.
- Support your discussion with references to relevant case studies, industry reports, or academic literature.
- **Insights, Interpretation, and Recommendations:**
 - Based on the results from all modelling techniques and churn analyses, identify the key drivers of churn and explain their significance in Omni Company's business environment.
 - Interpret the outputs from different interpretability methods.
 - Discuss the insights gained from different modelling approaches (e.g., Logistic Regression vs. tree-based models) and from different churn analysis perspectives (e.g., pre-churn behaviour vs. post-churn return tendencies).
 - Explain how these interpretations inform practical business decisions, such as which customer segments to target for retention campaigns and which service improvements to prioritise.
 - Make specific, actionable recommendations for reducing churn, supported by evidence from the model results and interpretability analyses.

Note:

- **These tasks must be implemented using Python. You need to submit your Python code along with your report.**



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Required Format:

- a) 5,000 words excluding references and statistical summaries or programming codes.
- b) All external sources of information must be referenced.
- c) Font – Times New Roman size 12.
- d) Spacing – 1.5.
- **Referencing Style:** For this course, you should use the **Harvard referencing style**. This includes both in-text citations and a reference list at the end of your assignment.
- **In-Text Citations:** When you refer to or paraphrase someone else's work within your text, you must include an in-text citation. Here are a couple of examples:
 - **Book:** (Ladley, 2020)
 - **Journal Article:** (Khatri & Brown, 2010)
 - **Website:** (Data Governance Institute, 2023)
- **Reference List:** At the end of your assignment, include a full list of references with all the details of each source. Examples:
 - **Book:** Ladley, J. (2020) *Data Governance: How to Design, Deploy, and Sustain an Effective Data Governance Program*. 2nd ed. Waltham: Elsevier.
 - **Journal Article:** Khatri, V. and Brown, C.V., 2010. Designing data governance. *Communications of the ACM*, 53(1), pp.148-152.
 - **Website:** Data Governance Institute (2023) *The Data Governance Framework*. Available at: <https://www.datagovernance.com/framework> (Accessed: 15 August 2024).

Submission & Formatting Guidelines:

When submitting your group assignment, please ensure that your file is properly formatted and named according to the following guidelines:

- **File Name Format:**
 - IS6052_ For group projects: **IS6052_GroupProject_GroupName**
 - Example: IS6052_GroupProject_Group1
- **File Type:**



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- Please submit your work as a **PDF** or **Word document (.docx)** along with the **Python file** (PyCharm or Jupyter)
- **File Content:**
 - **Title Page:** Include a title page with the following information:
 - Module Code: IS6052
 - Module Title: Predictive Analytics
 - Group Number
 - Student ID(s)
 - First and last names
 - Submission Date

Footers: Include page numbers in the footer of each page.

- **Submit your group assignment as a Word/PDF document along with the Python file (PyCharm or Jupyter) through Canvas.**
- **All group assignments will be checked for plagiarism using Turnitin.**
- **All group assignments will be graded according to the rubric provided in Table1.**

Table 1: IS6052 Individual Project Marking Rubric

Criteria	Excellent (85-100%)	Good (70-84%)	Satisfactory (50-69%)	Needs Improvement (30-49%)	Unsatisfactory (0-29%)
Dataset Description (15%)	Thorough and insightful description of the dataset, including detailed exploration and relevant context.	Good description of the dataset with adequate exploration and context provided.	Basic description of the dataset, with limited exploration and context.	Incomplete or unclear description of the dataset with minimal exploration or context.	No or very poor description of the dataset with little to no exploration or context.
Data Preparation and Feature Engineering (10%)	Comprehensive data cleansing with advanced feature engineering. All steps are well-documented and justified.	Good data cleansing with some feature engineering. Most steps are documented and justified.	Adequate data cleansing with basic feature engineering. Documentation is sufficient but lacks detail in some areas.	Minimal data cleansing and feature engineering. Poor documentation with little justification of steps.	Little to no data preparation or feature engineering. Documentation is incomplete or missing.
Predictive Analysis	Applies a variety of predictive techniques	Applies a variety of predictive	Applies predictive techniques with	Applies limited predictive	Fails to apply predictive



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(25%)	with excellent justification. Models are well-tuned, and results are robust.	techniques with good justification. Models are reasonably tuned with satisfactory results.	some justification. Models are basic with acceptable results but lack tuning.	techniques with poor justification. Models are poorly executed and produce weak results.	techniques effectively. Models are incorrect or produce inaccurate results.
Outlier Detection and Handling (15%)	Outliers are expertly identified and managed, with a strong impact on model performance. Methods are well-explained.	Outliers are identified and managed with noticeable impact on model performance. Methods are explained but lack some detail.	Outliers are addressed, but methods are basic and have limited impact on model performance. Explanation is adequate.	Outlier handling is minimal or ineffective. Methods have little to no impact on model performance. Explanation is unclear.	Outliers are not identified or handled, leading to significant issues with model performance. Methods are poorly explained.
Results, Evaluation, and Discussion, interpretations, recommendations (25%)	Results are presented clearly with excellent visualizations. Critical discussion is insightful, well-supported by literature. Python code is submitted.	Results are well-presented with good visualizations. Discussion is solid and supported by relevant literature.	Results are presented with basic visualizations. Discussion is satisfactory but lacks depth or support from literature.	Results are poorly presented with inadequate visualizations. Discussion is shallow and lacks support from relevant literature.	Results are unclear or incorrect with no visualizations. Discussion is minimal, lacking critical evaluation or literature. Python code is not submitted.
Formatting and Referencing (5%)	All formatting guidelines are followed precisely. Referencing is thorough and correctly formatted.	Most formatting guidelines are followed with few errors. Referencing is mostly complete and correctly formatted.	Formatting guidelines are followed but with several errors. Referencing is incomplete or incorrectly formatted in some parts.	Formatting guidelines are poorly followed. Referencing is inadequate or contains many errors.	Formatting guidelines are not followed. Referencing is missing or incorrect.
AI supervisory skills (5%)	Demonstrates exceptional ability to supervise AI tools effectively. Critically evaluates and refines AI-generated content, ensuring high-quality outputs. Shares detailed code and reports with clear	Shows good ability to supervise AI tools. Usually evaluates and refines AI suggestions but may miss minor issues. Shares code and reports with evidence of AI oversight.	Demonstrates basic AI supervision skills. Occasionally accepts AI output without critical evaluation. Shares code and reports but lacks detailed supervision.	Shows limited ability to supervise AI tools. Often accepts AI output without critical evaluation. Code and reports lack evidence of effective AI oversight.	Fails to supervise AI tools effectively. Blindly follows AI output without evaluation. Does not share code or reports or shows no evidence of AI oversight.



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	evidence of AI supervision.				
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AI Supervisory Skills:

Note: If you choose to use **AI tools like ChatGPT** to assist with your assignment, you must provide transparency in your work. Specifically, if you incorporate any content generated by ChatGPT or similar AI tools, you are required to share the link to the specific conversation or session with me along with your assignment submission. This ensures that all sources of information are properly acknowledged, and it helps maintain the integrity of your work. Failure to disclose the use of such tools may be considered a breach of academic honesty policies.

- As you interact with AI tools like ChatGPT, your task is not only to use the tool but to critically supervise the suggestions it provides. This means evaluating the accuracy, relevance, and applicability of AI-generated content. You should refine and adapt the AI's suggestions to better suit your assignment's context.
- You are required to share the code and the final report with clear documentation on how you used AI assistance. Your documentation should explain how you supervised the AI, what decisions you made to modify or accept its suggestions, and why.
- Your ability to supervise the AI will be evaluated based on how well you refine and adapt the AI-generated content. You will be assessed on your critical thinking, decision-making, and the quality of your final outputs. The emphasis will be on ensuring that you use AI as a tool to enhance your work, not as a crutch to produce uncritical or unrefined results.

Feedback Mechanisms

INTERIM FEEDBACK. Project groups may request feedback on progress throughout the course of the project via the following mechanisms: (1) Scheduled meeting with lecturer or (2) Email feedback on draft submissions.

Penalties

Submission of a project indicates that this is the student's own work. If the submission is found not to be original work, a mark of 0% will be automatically awarded. This project is



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worth 50% of the total module percentage for IS6052. Where work is submitted up to and including 7 days late, 10% of the total marks available shall be deducted from the mark achieved. Where work is submitted up to and including 14 days late, 20% of the total marks available shall be deducted from the mark achieved. Work submitted 15 days late or more shall not be accepted.

This project outline adheres (as appropriate) to the policies and codes of conduct outlined in Handbook Governing Curriculum Approval, the Policies and Guidelines Governing Academic Programmes, DMIS Guidelines, the Guide to Examinations for Staff and Students, the Plagiarism Policy, the Duty of Respect and Right to Dignity Policy, and the UCC Student Rules etc.